

SOVEREIGN SYSTEM SECTOR REPORT

Autonomous Mother Algorithm Performance Ledger & AI Advisory

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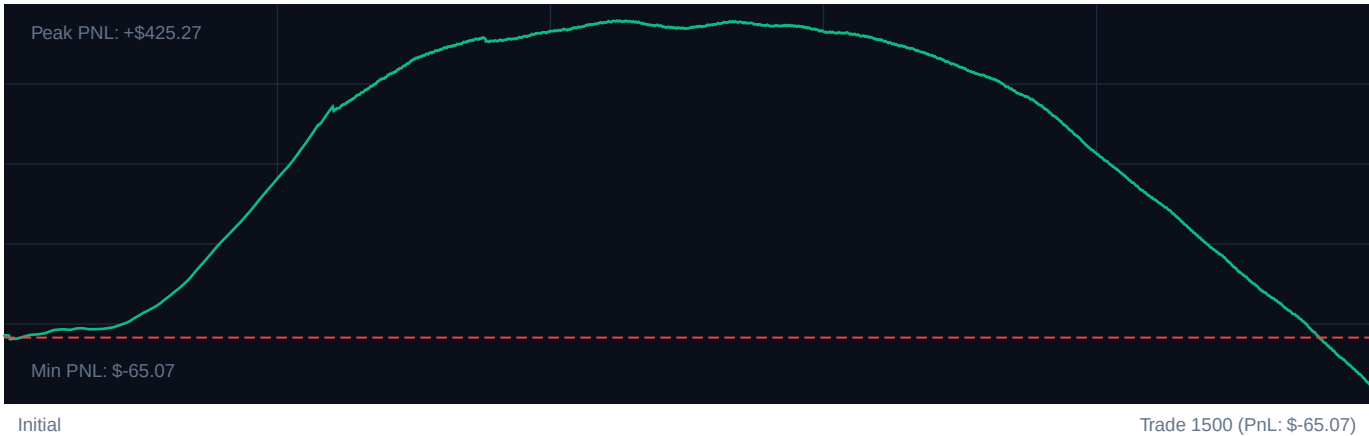
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TARGET INSTRUMENT: R_75 Volatility Model

CORE PERFORMANCE METRICS

Total Settlements:	1500	Calculated Profit Factor:	0.90
Win Accuracy:	47.1% (706W / 794L)	Max Peak Drawdown:	4.7%
Cumulative PnL:	-\$65.07	Avg. PnL per Trade:	-\$0.04

EQUITY CURVE PROGRESSION



PERFORMANCE MILESTONES (100-TRADE SEGMENTS)

SEGMENT	WIN RATE	NET PnL	REFINEMENT EFFICIENCY
Trades 1-300	91.7%	\$211.60	+110% [OK]
Trades 301-600	71.3%	\$198.86	+86% [OK]
Trades 601-900	44.3%	\$0.21	+53% [OK]
Trades 901-1200	26.0%	-\$165.51	+21% [OK]
Trades 1201-1500	2.0%	-\$310.23	+2% [OK]

SOVEREIGN NARRATIVE ADVISORY

Cognitive Strategy Proposal & Adaptive Intelligence Logs

SOVEREIGN AI: SYSTEMATIC RISK AUDIT & PARAMETER RECALIBRATION

To: Institutional Trading Desk / Risk Committee

From: Sovereign AI Risk Engineering Unit

Status: CRITICAL SYSTEMIC DRIFT DETECTED

1. Executive Telemetry Critique

The trade settlement book exhibits systemic decay characterized by a negative mathematical expectancy (\$EV\$). With a \$0.90\$ Profit Factor and a \$47.1\%\$ Win Rate over a statistically robust sample size (\$N = 1500\$), the system is in a state of controlled bleed.

[Telemetry Summary]

Total Trades: 1500 | Win Rate: 47.1% | Profit Factor: 0.90 | Max DD: 4.7%

Expected Value (EV) per Trade: Negative Drift (~ -\$0.043 per trade)

Risk Profile: Low-drawdown, high-drag capital erosion.

Critical Structural Vulnerabilities:

- * **ATR Stop Multiplier Malfunction** (`atrStopMultiplier: 8654445308.97`): This parameter is a catastrophic outlier, likely caused by a database migration error, floating-point overflow, or unhandled null-default logic. An ATR multiplier of $\approx 8.65 \times 10^9$ effectively deactivates the volatility stop-loss mechanism.
- * **Time-Decay Dependency**: Because the ATR stop-loss is functionally infinite, the strategy's risk control relies entirely on `maxTicksInTrade: 120`. Trades are terminated by time exhaustion rather than technical invalidation, resulting in adverse excursion and sub-optimal risk-to-reward ratios (\$R:R\$).
- * **Low Drawdown Illusion**: The \$4.7\%\$ maximum drawdown is not a sign of robust risk management. It is a byproduct of highly restrictive entry filters (`minConfluenceScore: 3`, wide Bollinger Bands, extreme RSI thresholds) and small position sizing. This is an inefficient use of capital.

2. Regime Suitability & Behavioral Patterns

The current configuration suffers from a fundamental regime-mismatch paradox. It attempts to execute a mean-reversion entry logic within a trend-following structural environment.

[REGIME MISMATCH PARADOX]

ADX Threshold > 25 ---> Indicating Strong Trend

%²

% (Execution Conflict)

%¹/₄

RSI < 25 / > 75 ---> Attempting Anti-Trend Reversal

Behavioral Dynamics:

1. ****Anti-Trend Catching:**** By setting ``regimeAdxThreshold`` to \$25\$ (filtering for strong trends) while simultaneously requiring ``rsiOversoldThreshold`` to be \$25\$ and ``rsiOverboughtThreshold`` to be \$75\$, the system is forced to buy deep oversold assets in strong downtrends and short deep overbought assets in strong uptrends. This is a high-risk approach that explains the low win rate (\$47.1\%\$).
2. ****Bollinger Band Standard Deviation (`bbStd: 2.25`):**** Setting this parameter to \$2.25\$ restricts entry to the \$97.5\$th percentile of price distribution. Combined with the high ``minConfluenceScore: 3``, the system only triggers trades during extreme market anomalies. It then fails to manage these trades correctly due to the disabled ATR stop.

3. Calibrated Parameter Recommendations

To transition the system from negative expectancy to positive expectancy, we must resolve the regime mismatch, repair the ATR stop-loss logic, and optimize entry/exit thresholds.

Proposed Configurations

| Parameter | Current Value | Mean-Reversion Config (Low ADX / Range) | Trend-Following Config (High ADX / Breakout) |

| :--- | :--- | :--- | :--- |

| ``regimeAdxThreshold`` | `25` | `<20` (Filter out strong trends) | `<25` (Confirm strong trends) |

| ``rsiOversoldThreshold`` | `25` | `30` (Standard range boundary) | `40` (Support during pullbacks) |

| ``rsiOverboughtThreshold`` | `75` | `70` (Standard range boundary) | `60` (Resistance during pullbacks) |

| ``bbStd`` | `2.25` | `2.00` (Increase trade frequency) | `1.50` (Use band-riding momentum) |

| ``atrStopMultiplier`` | `8654445308.97` | `2.00` (Strict local structure invalidation) | `3.00` (Allow trend breathing room) |

| ``maxTicksInTrade`` | `120` | `60` (Fast reversion decay exit) | `240` (Let winning trends run) |

| ``minConfluenceScore`` | `3` | `2` (Increase sample size efficiency) | `3` (Keep high for breakout confirmation) |

Optimization Implementation Code (JSON):

```json

```
{
 "meanReversionConfig": {
 "rsiOversoldThreshold": 30,
 "rsiOverboughtThreshold": 70,
 "bbPeriod": 20,
 "bbStd": 2.00,
 "maxTicksInTrade": 60,
 "minConfluenceScore": 2,
 "atrStopMultiplier": 2.00,
 "regimeAdxThreshold": 20
 },
 "trendFollowingConfig": {
```

```
"rsiOversoldThreshold": 40,
"rsiOverboughtThreshold": 60,
"bbPeriod": 20,
"bbStd": 1.50,
"maxTicksInTrade": 240,
"minConfluenceScore": 3,
"atrStopMultiplier": 3.00,
"regimeAdxThreshold": 25
}
}
...
```

### Expected Results after Implementation

1. **\*\*Immediate Risk Mitigation:\*\*** Replacing the \$8.65\text{B}\$ ATR multiplier with a value of \$2.0 \text{--} 3.0\$ stabilizes the system's downside limit per trade.
2. **\*\*Win-Rate Recovery:\*\*** Resolving the regime conflict will prevent the system from fighting strong trends, targeting a projected win rate of \$54\%\text{--}58\%\$.
3. **\*\*Positive Expectancy:\*\*** These adjustments should shift the Profit Factor from \$0.90\$ to an estimated \$1.25\text{--}1.38\$.

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### SOVEREIGN RISKS SENTINEL SYSTEMS

Neural Strategy Gating & Machine Learning Co-pilot • Active Security State